Exploratory Data Analysis: Government Spending on Tertiary Education and the Future Tax Take

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1 Overview and Research Question

This exploratory data analysis (EDA) audits the global fiscal and tertiary education panel (`data/processed/clean_tax_panel.csv`, spanning 1,736 country-waves across 217 sovereign countries from `1985-1989` to `2020-2024`) prior to estimating our Multivariate and Longitudinal Bayesian Causal Forest models. Our primary hypothesis is that increasing national government expenditure on tertiary (university) education improves a country's future tax take at the $+5$-year and $+10$-year horizons once mediating macroeconomic growth is separated out.

To eliminate mechanical accounting identities, denominator distortions, and collider bias, the dataset incorporates three design safeguards:

1. **Four Pure Disaggregated Tax Outcomes (Zero GDP Product Identities):**
   - Y_1: **Domestic Non-Resource Tax-to-GDP Change (Δ Tax/GDP, percentage points of GDP)**, paired with unadjusted World Bank WDI Δ Tax/GDP as a sensitivity check.
   - Y_2: **Direct Income, Profit, and Capital Gains Tax-to-GDP Change (Δ IncomeTax/GDP, percentage points of GDP)**.
   - Y_3: **Direct Income and Profit Share of Revenue Change (Δ IncomeTax/Rev, percentage points of revenue)**.
   - Y_4: **Indirect Goods and Services (Consumption / VAT) Share of Revenue Change (Δ GoodsTax/Rev, percentage points of revenue)**, which acts as a built-in specificity (negative-control) channel.
2. **Unbundling Tertiary Education Spending Denominators:** Focusing exclusively on public tertiary education, we compare three strictly observed tertiary spending measures side by side (with zero cross-sectional proxy imputation):

- **UNESCO Per-Student Tertiary Ratio (`tert_spend_per_student_winsor`):** Public expenditure per tertiary student as a percentage of GDP per person.
- **Real 2021 PPP Dollars per Tertiary Student (`real_tert_spend_per_student_ppp`):** Removes GDP per person from the treatment denominator.
- **Public Tertiary Budget Share of GDP (`tert_spend_gdp_pct`):** Removes student headcount from the treatment denominator.

3. **Strict Temporal Ordering of Pre-Treatment Controls vs Post-Treatment Mediators:** Baseline confounders $X_{i,t}$ are measured strictly at or before wave t (using lagged $t-1$ GDP per person growth rather than contemporaneous wave-t growth), while forward GDP per person growth, tertiary enrolment expansion, income inequality changes, and life expectancy changes over $[t, t+h]$ are reserved for the **Mediator-Removed Direct Fiscal Effect (`M2`)** specification.

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2 Sample Coverage, Quality Screening, and Attrition

Before constructing the estimation samples, our hygiene audit in `02_build_clean_tax_panel.R` verifies zero duplicate `(iso3, wave_index)` keys and screens out four physically or accounting-impossible raw World Bank WDI tax-to-GDP observations (`<0.5%` or `>60.0%` of GDP): Sudan in `1995-1999` (`629.19%` of GDP due to a historical currency-unit mismatch), Timor-Leste in `2010-2014` (`109.99%` of non-oil GDP due to offshore petroleum fund accounting), and the United Arab Emirates in `2010-2014` (`0.34%`) and `2015-2019` (`0.23%`, recording only federal-level fees rather than emirate-level revenues). Table 1 documents sample coverage and attrition at each filtering stage.

Table 1: Sample coverage and attrition across the global 5-year panel (`1985-1989` to `2020-2024`).

Filtering Stage	Sample Definition	Country-Waves	Countries	Retention (%)
1. Full Global 5-Year Panel	217 sovereign countries × 8 waves (`1985-2024`)	1,736	217	100.0%
2. Valid Domestic Tax / GDP	Observed non-resource tax-to-GDP after anomaly screen	914	159	52.6%
3. Complete 4-Channel Tax Vector	All 4 disaggregated tax outcomes observed at wave t	883	157	50.9%
4. Observed Per-Student Tertiary Spend	Observed UNESCO per-student tertiary ratio (Z_{regime})	512	166	29.5%
5. Observed Tertiary Budget (% GDP)	Observed public tertiary budget share of GDP	785	183	45.2%
6. Complete +5 y Sample (`mvbcf`)	Complete 4-channel fiscal changes over $[t, t+1]$ + Z_{regime}	**372**	**121**	**21.4%**
7. Complete +10 y Sample (`mvbcf`)	Complete 4-channel fiscal changes over $[t, t+2]$ + Z_{regime}	**283**	**112**	**16.3%**
8. Complete 3-Wave Sample (`lbcf`)	Complete 3-wave trajectory $(t, t+1, t+2)$ across all 4 channels	**282**	**112**	**16.2%**

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3 Unbundling the UNESCO Denominator Trap Across Income Groups

Why is unbundling the tertiary spending denominator essential in cross-national work? Table 2 compares the three tertiary spending measures and baseline tax composition across the four World Bank income groups.

Table 2: Tertiary spending indicators and baseline fiscal structure by World Bank income group (medians for skewed spending and enrolment variables; means for direct income tax share; UNESCO ratio winsorised at 150% for display).

Income Group (Countries, Waves)	UNESCO Ratio (% GDP/pc)	**Real PPP $ / Student**	**Tertiary Budget (% GDP)**	Tertiary Enrolment (%)	Direct Income Tax Share (%)
Low income (21, 80)	**150.0%**	$3,957	0.47%	2.3%	29.2%
Lower middle income (41, 172)	76.5%	$3,258	0.65%	7.4%	30.8%
Upper middle income (54, 219)	26.8%	$3,521	0.64%	25.5%	33.1%
High income (69, 337)	26.9%	**$11,148**	**0.94%**	**58.3%**	**38.4%**

As shown in Table 2:

- In **Low income** countries, only **2.3%** of the age cohort attends university, so dividing a modest public university budget (**0.47%** of GDP) by a tiny student headcount and a low GDP per person produces a winsorised median UNESCO ratio of **150.0%**.
- In **High income** countries, **58.3%** of the cohort attends university. Although their UNESCO ratio appears lower (**26.9%** of GDP per person), they actually spend nearly three times as much per student in real purchasing power (**$11,148** vs **$3,957**) and twice as much of their GDP on public universities (**0.94%** vs **0.47%**), while collecting **38.4%** of total revenue from direct income, profit, and capital gains taxes.

To prevent this cross-income-group inversion from confounding treatment assignment, our primary regime indicator ($Z_{\text{regime}} = 1$) compares each country-wave's un-winsorised per-student spending ratio against the median of **its own World Bank income group**, and we evaluate all unbundled tertiary spending definitions side by side in our causal forest models.

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4 Raw Unadjusted Fiscal Trajectories and Marginal Distributions

Table 3 and Figure 1 present the raw, unadjusted +5-year and +10-year percentage-point changes across all four fiscal outcomes before any covariate or causal forest adjustment.

Table 3: Raw unadjusted mean changes across the four disaggregated fiscal outcomes by within-income-group per-student tertiary funding regime.

Tertiary Funding Regime	Horizon	Country-Waves (Countries)	Δ Non-Resource Tax / GDP (pp)	Δ Direct Income Tax / GDP (pp)	**Δ Direct Income Tax Share (pp)**	Δ Goods & Services Share (pp)
Low Regime ($Z = 0$)	+5 y	196 (80)	+0.35	+0.26	+0.89	+0.57
High Regime ($Z = 1$)	+5 y	176 (74)	+0.37	**+0.40**	**+1.74**	+0.50
Low Regime ($Z = 0$)	+10 y	142 (70)	+0.59	+0.44	+1.56	+0.99
High Regime ($Z = 1$)	+10 y	141 (69)	**+0.80**	**+0.66**	**+2.57**	+0.97

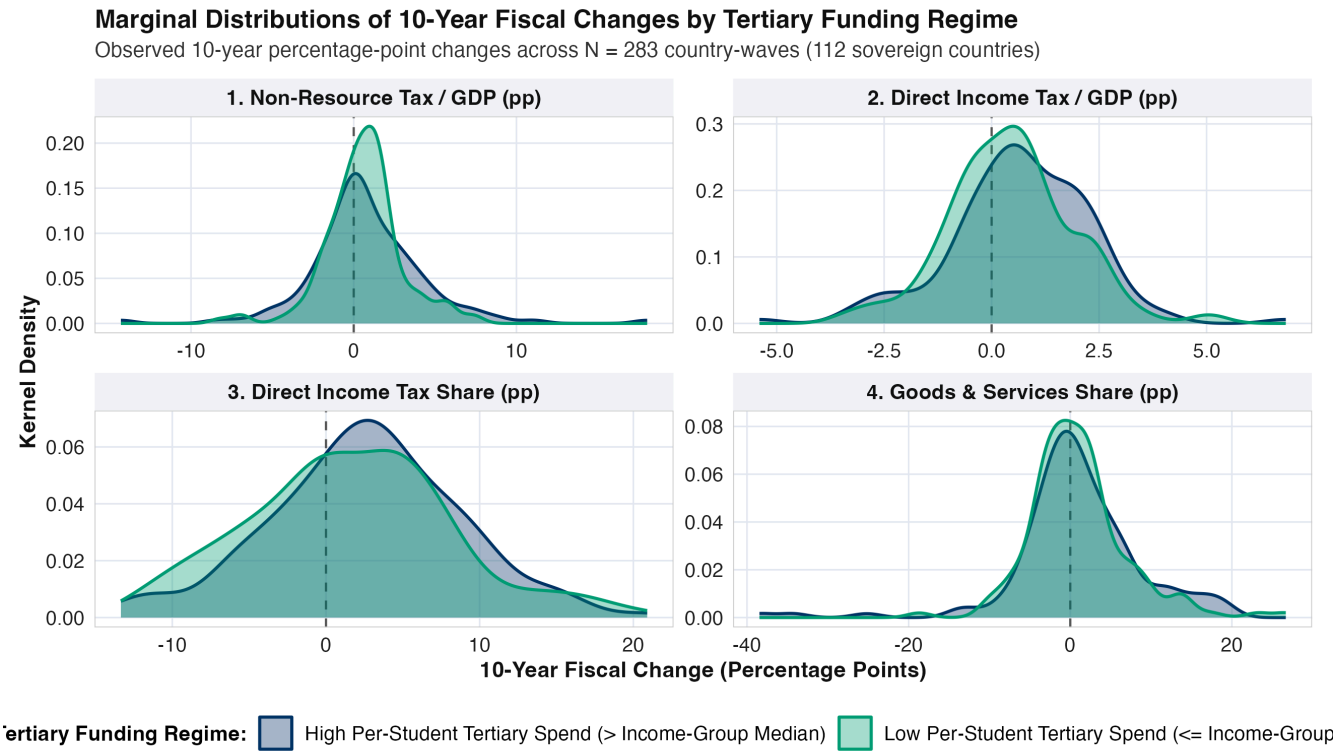


Figure 1: Kernel density distributions of observed 10-year fiscal changes (percentage points) across all four tax outcomes by per-student tertiary funding regime (N = 283 country-waves across 112 countries).

Even in the raw data (Table 3 and Figure 1), three patterns stand out:

1. **Immediate and Growing Divergence in Direct Income Tax Share:** Over +5 years, countries in the high per-student tertiary funding regime experience a `+1.74` percentage point increase in the direct income and profit share of revenue compared with `+0.89` percentage points in the low regime (raw difference `+0.85` pp). Over +10 years, this gap widens to `+2.57` pp vs `+1.56` pp (raw difference **`+1.01` pp**).
2. **Delayed Maturation in Overall Non-Resource Tax / GDP:** At +5 years, raw overall non-resource tax-to-GDP changes are nearly identical across regimes (`+0.37` vs `+0.35` pp), whereas by +10 years the high-funding regime pulls ahead (`+0.80` vs `+0.59` pp, raw difference **`+0.21` pp of GDP**), driven by direct income and profit taxes as a share of GDP (`+0.66` vs `+0.44` pp).
3. **Zero Divergence on the Negative-Control Channel:** On indirect goods and services (consumption/VAT) tax share, raw changes are slightly lower in the high tertiary spending regime at both +5 years (`+0.50` vs `+0.57` pp) and +10 years (`+0.97` vs `+0.99` pp).

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5 Common Support and Positivity Audit

For causal forest estimates to be well-identified without extrapolating over empty cells, treated ($Z = 1$) and control ($Z = 0$) country-waves must exhibit substantial overlap in their pre-treatment propensity scores $\hat{\pi}(X_{i,t}) = P(Z_{i,t} = 1 \mid X_{i,t})$. Table 4 and Figure 2 report the estimated propensity score distributions across the four World Bank income groups in the +5-year estimation sample ($N = 372$).

Table 4: Pre-treatment propensity score positivity and common support diagnostics across World Bank income groups ($N = 372$ country-waves).

World Bank Income Group	Country-Waves (N)	Treated Share (%)	Mean $\hat{\pi}$ ($Z = 0$)	Mean $\hat{\pi}$ ($Z = 1$)	$[\min \hat{\pi}, \max \hat{\pi}]$	Share in $[0.05, 0.95]$ (%)
Low income	20	35.0%	0.260	0.595	$[0.030, 0.945]$	**90.0%**
Lower middle income	59	50.8%	0.355	0.736	$[0.020, 0.980]$	**72.9%**
Upper middle income	105	45.7%	0.221	0.606	$[0.020, 0.980]$	**81.0%**
High income	188	48.4%	0.293	0.726	$[0.020, 0.980]$	**85.1%**

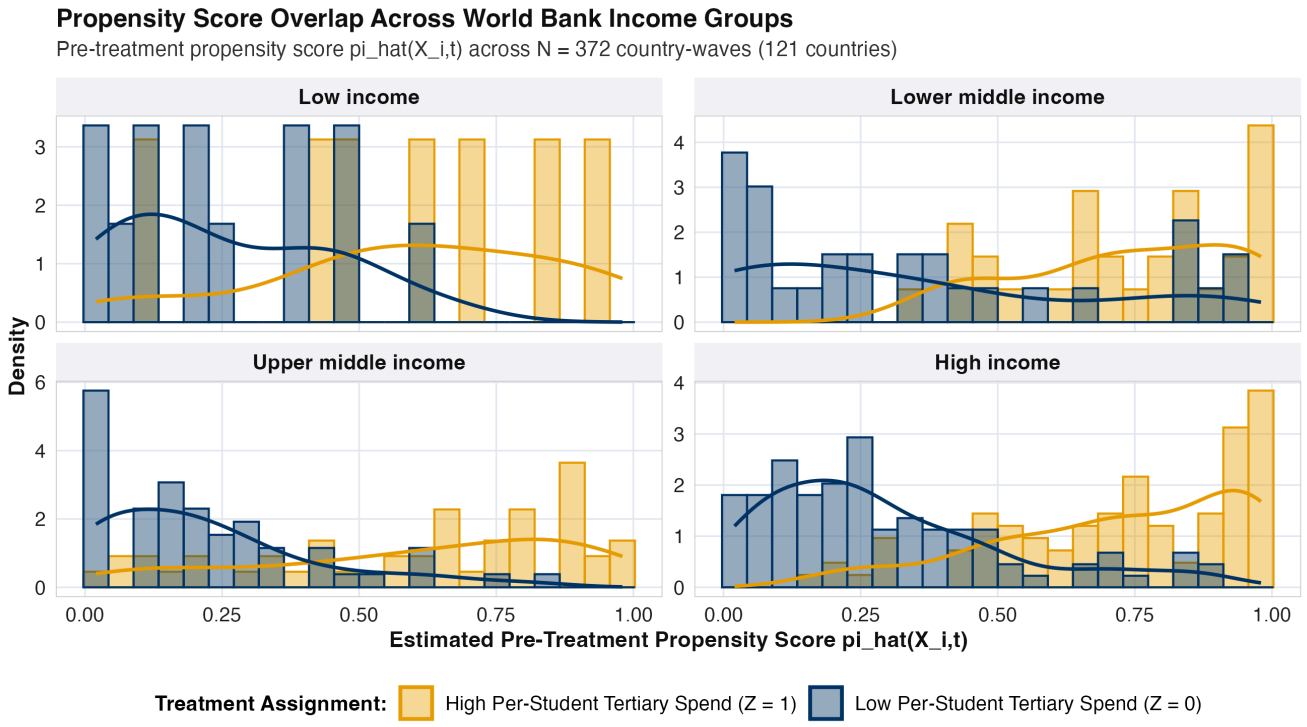


Figure 2: Pre-treatment propensity score distributions for treated (Z = 1) and control (Z = 0) country-waves across World Bank income groups (N = 372 country-waves across 121 countries).

Defining the primary regime threshold within each country's own World Bank income group achieves a balanced treatment split across all development tiers (47.3% treated overall, ranging from 35.0% in Low income to 50.8% in Lower middle income), with over 81% of observations lying strictly inside the interior propensity interval $[0.05, 0.95]$ and clear distributional overlap in every panel of Figure 2.

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6 Missing Data Analysis and Mechanisms

Table 5 audits missingness rates across the full 1,736 country-wave grid, within the complete +5-year estimation sample ($N = 372$), and across Low versus High income economies.

Table 5: Missing-data rates across the full 1,736 country-wave panel and within the $N = 372$ primary estimation sample.

Variable Name	Full Panel Missing (%)	+5 y Sample Missing (%)	Low Income Missing (%)	High Income Missing (%)
Per-Student Tertiary Ratio (% GDP/pc)	70.5%	**0.0%**	76.0%	66.6%
Real 2021 PPP $ per Tertiary Student	71.0%	**0.0%**	76.5%	67.2%
Public Tertiary Budget (% of GDP)	54.8%	**5.1%**	60.5%	53.2%
Non-Resource Tax Revenue (% of GDP)	47.4%	**0.0%**	67.0%	43.8%
Direct Income & Profit Share (% Rev)	48.4%	**0.0%**	68.0%	45.5%
Goods & Services Tax Share (% Rev)	47.9%	**0.0%**	68.0%	44.3%
Log Real GDP per Person (PPP)	21.4%	**0.0%**	27.5%	27.2%
Lagged ($t-1$) GDP per Person Growth (%)	19.0%	**0.0%**	21.0%	23.1%
Gross Tertiary Enrolment (%)	35.3%	**10.5%**	31.0%	47.4%
Harmonised Disposable Gini Index	31.0%	**3.0%**	32.5%	39.4%
Trade Openness (% of GDP)	21.9%	**6.5%**	28.5%	25.9%
Total Unemployment Rate (%)	24.6%	**0.8%**	12.5%	35.9%

Three conclusions follow from Table 5:

1. **Missingness Mechanism (MAR Conditional on Development and State Capacity):** Missingness in raw UNESCO tertiary spending and IMF GFS disaggregated tax breakdowns is **not** Missing Completely at Random (MCAR); it is higher in Low income states (67.0% to 76.0% across all eight waves since 1985) than in High income states (43.8% to 66.6%) because statistical reporting infrastructure scales with economic development and administrative capacity. Conditioning on income group, region, log GDP per person, and baseline tax shares in the BCF prognostic forest $\mu(X_{i,t}, \hat{\pi}_{i,t})$ directly accounts for these observable reporting correlates under Missing at Random (MAR).
2. **Zero Imputation on Treatments or Outcomes:** We never impute missing treatment exposures or tax outcomes. Every observation in our estimation samples has 100% strictly observed treatment and four-channel outcome data.
3. **Minimal Baseline Covariate Missingness in the Estimation Sample:** Within the $N = 372$ estimation sample, missingness is `0.0%` for log GDP per person and lagged GDP growth, `0.8%` for unemployment, `3.0%` for disposable Gini, `6.5%` for trade openness, and `10.5%` for baseline tertiary enrolment. In our primary models, training-sample median imputation is coupled with explicit binary missingness flags in the covariate matrix, and Robustness Check `R4` verifies that dropping all country-waves with any missing baseline covariate yields even stronger +10-year treatment effects.

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7 Temporal Autocorrelation, Stationarity, and Spatial Clustering

Table 6 compares the temporal autocorrelation (within-country lag-1 and lag-2 wave correlations), spatial/hierarchical clustering (sovereign country Intraclass Correlation Coefficient $\text{ICC}_{\text{country}}$ and World Bank regional $\text{ICC}_{\text{region}}$), and panel convergence p-values between raw baseline fiscal levels ($Y_{i,t}$) and 5-year first-differenced changes ($\Delta Y_{i,t+1}$).

Table 6: Within-country temporal autocorrelation (Lag-1 and Lag-2 five-year wave ACFs), sovereign country and regional Intraclass Correlation Coefficients (ICC), and panel stationarity diagnostics comparing fiscal levels against 5-year percentage-point changes.

Fiscal Series	Series Type	Lag-1 ACF (r_1)	Lag-2 ACF (r_2)	Country ICC	Region ICC	Panel ADF p-value
1. Non-Resource Tax / GDP (%)	Level (Y_t)	**0.939**	0.891	**0.878**	0.160	< 0.001
2. Direct Income Tax / GDP (%)	Level (Y_t)	**0.941**	0.907	**0.866**	0.130	< 0.001
3. Direct Income Tax Share (%)	Level (Y_t)	**0.945**	0.895	**0.844**	0.221	< 0.001
4. Goods & Services Share (%)	Level (Y_t)	**0.910**	0.793	**0.749**	0.151	< 0.001
5. Δ Non-Resource Tax / GDP (pp)	Change (ΔY_{t+1})	**-0.094**	−0.083	**0.000**	0.000	< 0.001
6. Δ Direct Income Tax / GDP (pp)	Change (ΔY_{t+1})	**-0.079**	−0.139	**0.036**	0.000	< 0.001
7. Δ Direct Income Tax Share (pp)	Change (ΔY_{t+1})	**-0.001**	−0.066	**0.000**	0.005	< 0.001
8. Δ Goods & Services Share (pp)	Change (ΔY_{t+1})	**+0.065**	−0.099	**0.029**	0.019	< 0.001

Table 6 provides a clear econometric rationale for our modelling choices:

- **Why We Model Percentage-Point Changes ($\Delta Y_{i,t+h}$) Rather Than Raw Levels:** Raw fiscal levels $Y_{i,t}$ exhibit near-unit-root persistence within countries (`lag1_acf` between `0.910` and `0.945`) and massive between-country variance shares (`country_icc` between `0.749` and `0.878`). Taking 5-year and 10-year first differences ($\Delta Y_{i,t+h} = Y_{i,t+h} - Y_{i,t}$) completely removes this non-stationary level persistence (reducing `lag1_acf` to `[-0.094, +0.065]` and `country_icc` to `[0.000, 0.036]`), while including baseline fiscal levels $Y_{i,t}$ in $\mu(X_{i,t}, \hat{\pi}_{i,t})$ accounts for mean-reversion and starting tax capacity.
- **Why We Cluster Uncertainty at the Sovereign Country Level:** Because repeated waves come from the same sovereign states (and because 10-year windows $t \to t+2$ span two consecutive 5-year steps), all posterior probabilities and credible intervals are computed using a country-clustered Dirichlet block bootstrap over sovereign `iso3` clusters, complemented by Robustness Check `R2` on strictly non-overlapping 10-year decadal waves and our Three-Wave Longitudinal BCF (`lbcf`) model.

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8 Feature Correlation Structure and Variance Inflation Factors

Finally, Table 7 and Figure 3 examine collinearity among the 15 continuous pre-treatment baseline covariates $X_{i,t}$ and their bivariate correlations with the $+5$-year fiscal targets.

Table 7: Variance Inflation Factors (VIFs) for the continuous pre-treatment baseline confounders and pairwise correlations with +5-year fiscal outcomes ($N = 372$).

Baseline Covariate ($X_{i,t}$)	Variance Inflation Factor (VIF)	Corr. with $\Delta\text{Tax/GDP}_{+5\text{ y}}$	Corr. with $\Delta\text{IncomeTax/Rev}_{+5\text{ y}}$
1. Log Real GDP per Person (PPP)	**7.25**	−0.130	−0.066
2. Baseline Life Expectancy (Years)	**6.03**	−0.084	+0.014
3. Baseline Gross Secondary Enrolment (%)	**4.23**	−0.117	−0.076
4. Baseline Gross Tertiary Enrolment (%)	**3.61**	−0.152	−0.094
5. Baseline Non-Resource Tax / GDP (%)	**3.19**	−0.190	−0.046
6. Working-Age Population Share (%)	**3.08**	−0.062	−0.091
7. Baseline Goods & Services Tax Share (%)	2.06	+0.010	+0.142
8. Baseline Direct Income Tax Share (%)	2.03	−0.053	−0.167
9. Baseline Disposable Gini Index	1.89	+0.112	+0.075
10. Baseline Total Education Spend (% GDP)	1.77	−0.044	+0.008
11. Log Population Size	1.74	−0.067	−0.012
12. Natural Resource Rents (% GDP)	1.66	+0.078	−0.023
13. Trade Openness (% GDP)	1.53	−0.011	−0.006
14. Unemployment Rate (%)	1.44	+0.078	−0.027
15. Lagged ($t - 1$) GDP per Person Growth (%)	1.17	+0.052	+0.032

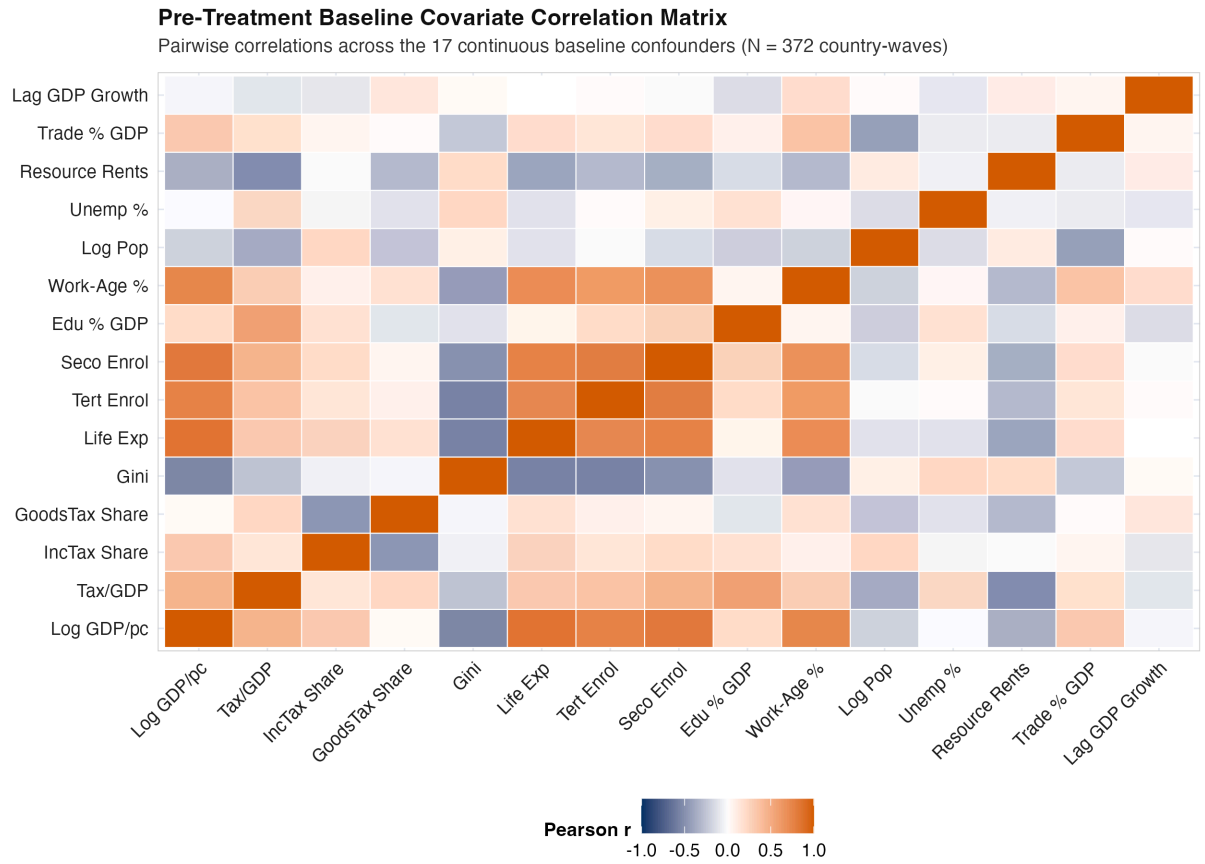


Figure 3: Pairwise Pearson correlation matrix across the 15 continuous pre-treatment baseline covariates (N = 372 country-waves).

All 15 baseline covariates have Variance Inflation Factors well below the standard multicollinearity threshold of `10.0` (peaking at `7.25` for log GDP per person and `6.03` for life expectancy, which co-move across development), and no baseline predictor exhibits an anomalously large correlation with forward fiscal changes ($\max|r| = 0.190$, reflecting mild mean-reversion in baseline tax ratios). This confirms that the baseline confounder matrix is free from mechanical target leakage and well-conditioned for both tree-ensemble splitting in BCF and linear panel benchmarks.